Decision Tree

Entropy, Information Gain and Step-by-Step Tree Construction

1 1. What is a Decision Tree?

A Decision Tree is a supervised machine learning algorithm used for classification and regression.

For classification, a Decision Tree works like a sequence of questions:

Ask a question → Split the data → Ask another question → Reach a decision

For example:

Is the weather Sunny, Overcast or Rain?

Depending on the answer, we move to another question or directly reach a final decision.

Key Concept

Basic idea:
A Decision Tree divides the dataset into smaller and more pure groups by selecting the best feature at each step.
The feature that gives the highest Information Gain is selected for the decision node.

2 2. Important Terms

- **Root Node:** The first decision node of the tree.
- **Decision Node:** A node where the dataset is split using a feature.
- **Branch:** The result of a decision or feature value.
- **Leaf Node:** The final prediction.
- **Splitting:** Dividing the dataset into smaller subsets.

Question: How do we decide which feature becomes the root node?

We calculate **Information Gain (IG)** for each candidate feature.

The feature with the highest Information Gain becomes the root node.

3 3. Algorithm for Decision Tree Classifier

1. Start with the complete dataset.
2. Count the positive and negative target values and write them as

   $$S\{+x, -y\}$$

   where $+x$ represents Yes values and $-y$ represents No values.
3. Calculate the entropy of the complete dataset.
4. For every candidate feature:
   - Split the dataset according to the feature values.
   - For every resulting subset, write its $S\{+x, -y\}$ values.
   - Calculate the entropy of every subset.
   - Calculate Information Gain.
5. Compare the Information Gain values.
6. Select the feature having the highest Information Gain.
7. Make that feature the root node or the next decision node.
8. Check each resulting branch:
   - If all records have the same class, make it a leaf node.
   - If the records are mixed, repeat the same process using the remaining features.
9. Continue until the branches become pure or another stopping condition is reached.

4 4. Entropy Calculation

Entropy measures the impurity or uncertainty in a group of data.

For a binary classification problem:

$$\boxed{S\{+x, -y\} = -\frac{x}{x+y}\log_2\left(\frac{x}{x+y}\right) - \frac{y}{x+y}\log_2\left(\frac{y}{x+y}\right)}$$

Key Concept

Special cases:
If a subset contains only Yes values:

$$S\{+x, -0\} = 0$$

If a subset contains only No values:

$$S\{+0, -y\} = 0$$

Therefore, a completely pure group has entropy 0.

5 5. Information Gain

Information Gain tells us how useful a feature is for splitting the dataset.

For a feature:

$$\boxed{IG = \text{Entropy of parent dataset} - [\text{proportion of subset}_1 \times \text{entropy of subset}_1 + \cdots]}$$

We calculate IG for all candidate features and select the feature with the highest IG.

Key Concept

Rule for selecting a node:

$$\boxed{\text{Highest Information Gain} \Rightarrow \text{Best split}}$$

The best split is the one that reduces impurity or uncertainty the most.

6 6. Example: Play Football

We want to predict whether a person will **Play Football** based on the weather conditions.

7 7. Step 1: Calculate Entropy of the Entire Dataset

First, count the target values.

$$\text{Yes} = 9$$

$$\text{No} = 5$$

Therefore:

Day	Outlook	Humidity	Wind	Play Football?
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

Table 1: Football Dataset

$$\boxed{S\{+9, -5\}}$$

Total records:

$$9 + 5 = 14$$

Now calculate entropy:

$$S\{+9, -5\} = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)$$

$$\boxed{S\{+9, -5\} = 0.940}$$

This is the impurity of the complete dataset before any split.

8 8. Step 2: Calculate IG for Outlook

Outlook has three possible values:

$$\text{Sunny, Overcast, Rain}$$

8.1 Sunny

Sunny records:

D1, D2, D8, D9, D11

$$\text{Yes} = 2, \qquad \text{No} = 3$$

Therefore:

$$S\{+2, -3\}$$

$$= -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)$$

$$\boxed{S\{+2, -3\} = 0.971}$$

8.2 Overcast

Overcast records:

D3, D7, D12, D13

$$\text{Yes} = 4, \qquad \text{No} = 0$$

Therefore:

$$S\{+4, -0\} = 0$$

8.3 Rain

Rain records:

D4, D5, D6, D10, D14

$$\text{Yes} = 3, \qquad \text{No} = 2$$

Therefore:

$$S\{+3, -2\}$$

$$= -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$$

$$\boxed{S\{+3, -2\} = 0.971}$$

8.4 Information Gain for Outlook

$$IG(Outlook) = 0.940 - \left[\frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971)\right]$$

$$\boxed{IG(Outlook) \approx 0.247}$$

9 9. Step 3: Calculate IG for Humidity

Humidity has two values:

$$\text{High, Normal}$$

9.1 High

High records:

D1, D2, D3, D4, D8, D12, D14

$$\text{Yes} = 3, \qquad \text{No} = 4$$

$$S\{+3, -4\}$$

$$= -\frac{3}{7}\log_2\left(\frac{3}{7}\right) - \frac{4}{7}\log_2\left(\frac{4}{7}\right)$$

$$\boxed{S\{+3, -4\} = 0.985}$$

9.2 Normal

Normal records:

D5, D6, D7, D9, D10, D11, D13

$$\text{Yes} = 6, \qquad \text{No} = 1$$

$$S\{+6, -1\}$$

$$= -\frac{6}{7}\log_2\left(\frac{6}{7}\right) - \frac{1}{7}\log_2\left(\frac{1}{7}\right)$$

$$\boxed{S\{+6, -1\} = 0.592}$$

9.3 Information Gain for Humidity

$$IG(Humidity) = 0.940 - \left[\frac{7}{14}(0.985) + \frac{7}{14}(0.592)\right]$$

$$\boxed{IG(Humidity) \approx 0.152}$$

10 10. Step 4: Calculate IG for Wind

Wind has two values:

$$\text{Weak, Strong}$$

10.1 Weak

Weak records:

D1, D3, D4, D5, D8, D9, D10, D13

$$\text{Yes} = 6, \qquad \text{No} = 2$$

$$S\{+6, -2\}$$

$$= -\frac{6}{8}\log_2\left(\frac{6}{8}\right) - \frac{2}{8}\log_2\left(\frac{2}{8}\right)$$

$$\boxed{S\{+6, -2\} = 0.811}$$

10.2 Strong

Strong records:

D2, D6, D7, D11, D12, D14

$$\text{Yes} = 3, \qquad \text{No} = 3$$

$$S\{+3, -3\}$$

$$= -\frac{3}{6}\log_2\left(\frac{3}{6}\right) - \frac{3}{6}\log_2\left(\frac{3}{6}\right)$$

$$\boxed{S\{+3, -3\} = 1}$$

10.3 Information Gain for Wind

$$IG(Wind) = 0.940 - \left[\frac{8}{14}(0.811) + \frac{6}{14}(1)\right]$$

$$\boxed{IG(Wind) \approx 0.048}$$

11 11. Step 5: Compare Information Gain Values

We have calculated IG for all candidate features.

Feature	Information Gain
Outlook	0.247
Humidity	0.152
Wind	0.048

The highest Information Gain is:

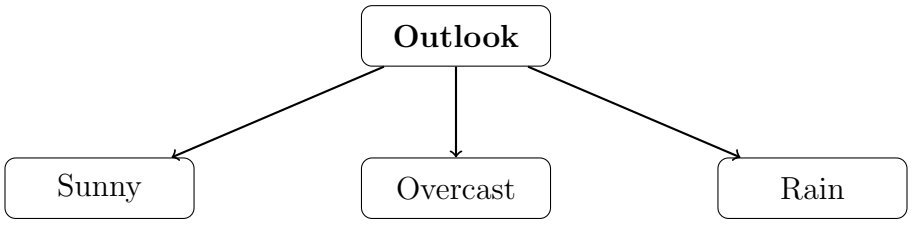
$$\boxed{IG(Outlook) = 0.247}$$

Therefore:

$$\boxed{\text{Outlook becomes the Root Node}}$$

12 12. Step 6: Construct the Root Node

Now we start constructing the tree.



Now examine each branch.

13 13. Step 7: Examine the Overcast Branch

For Overcast:

$$S\{+4, -0\}$$

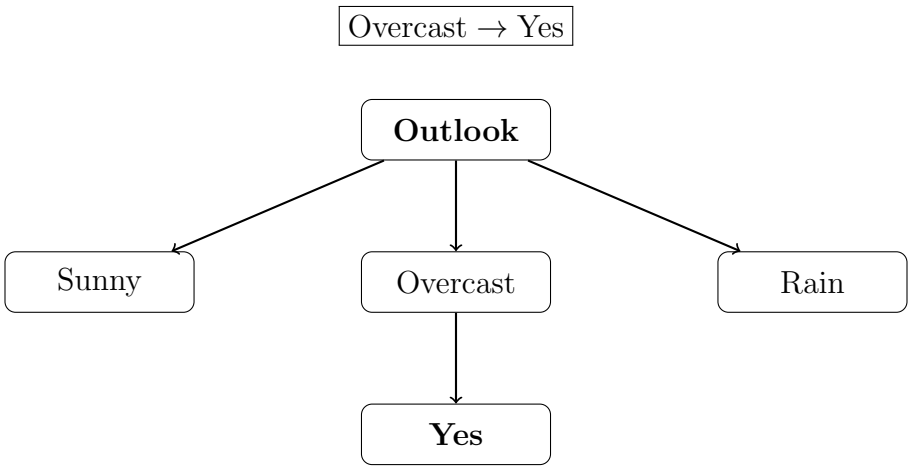
All records are Yes.

Therefore:

$$S\{+4, -0\} = 0$$

The branch is already pure.

Hence:



14 14. Step 8: Examine the Sunny Branch

For Sunny:

$$S\{+2, -3\}$$

The branch is not pure because it contains both Yes and No.

Therefore, we must split the Sunny subset again.

The remaining features are:

$$Humidity, \quad Wind$$

14.1 Split Sunny by Humidity

Sunny + High:

D1, D2, D8

$$S\{+0, -3\} = 0$$

Sunny + Normal:

D9, D11

$$S\{+2, -0\} = 0$$

Therefore:

$$IG(Humidity \mid Sunny) = 0.971 - \left[\frac{3}{5}(0) + \frac{2}{5}(0)\right]$$

$$\boxed{IG(Humidity \mid Sunny) = 0.971}$$

14.2 Split Sunny by Wind

Sunny + Weak:

D1, D8, D9

$$S\{+1, -2\} = 0.918$$

Sunny + Strong:

D2, D11

$$S\{+1, -1\} = 1$$

Therefore:

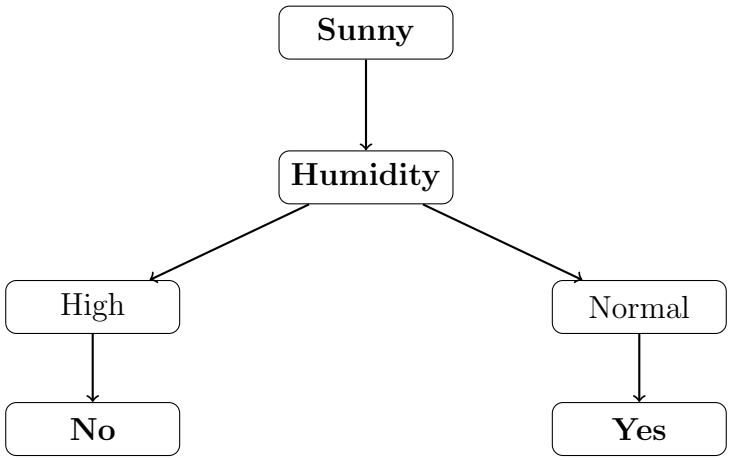
$$IG(Wind \mid Sunny) = 0.971 - \left[\frac{3}{5}(0.918) + \frac{2}{5}(1)\right]$$

$$\boxed{IG(Wind \mid Sunny) \approx 0.020}$$

Therefore:

$$\boxed{Humidity \text{ is selected for the Sunny branch}}$$

14.3 Construct the Sunny Node



Both branches are pure:

$$S\{+0, -3\} = 0$$

and

$$S\{+2, -0\} = 0$$

Therefore:

$$\boxed{Sunny + High \rightarrow No}$$

$$\boxed{Sunny + Normal \rightarrow Yes}$$

15 15. Step 9: Examine the Rain Branch

For Rain:

$$S\{+3, -2\}$$

This branch is not pure because it contains both Yes and No.

The remaining features are:

$$Humidity, \quad Wind$$

15.1 Split Rain by Humidity

Rain + High:

D4, D14

$$S\{+1, -1\} = 1$$

Rain + Normal:

D5, D6, D10

$$S\{+2, -1\} = 0.918$$

Therefore:

$$IG(Humidity \mid Rain) = 0.971 - \left[\frac{2}{5}(1) + \frac{3}{5}(0.918)\right]$$

$$\boxed{IG(Humidity \mid Rain) \approx 0.020}$$

15.2 Split Rain by Wind

Rain + Weak:

D4, D5, D10

$$S\{+3, -0\} = 0$$

Rain + Strong:

D6, D14

$$S\{+0, -2\} = 0$$

Therefore:

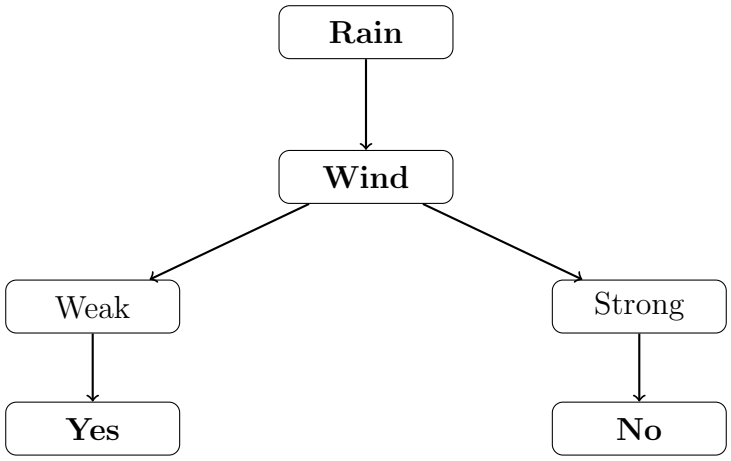
$$IG(Wind \mid Rain) = 0.971 - \left[\frac{3}{5}(0) + \frac{2}{5}(0)\right]$$

$$\boxed{IG(Wind \mid Rain) = 0.971}$$

Therefore:

$$\boxed{Wind \text{ is selected for the Rain branch}}$$

15.3 Construct the Rain Node



Both branches are pure:

$$S\{+3, -0\} = 0$$

and

$$S\{+0, -2\} = 0$$

Therefore:

$$\boxed{Rain + Weak \rightarrow Yes}$$

$$\boxed{Rain + Strong \rightarrow No}$$

16 16. Step-by-Step Tree Construction

The complete construction can be summarized as follows.

Step 1: Start with the complete dataset

$$S\{+9, -5\}$$

Calculate IG for:

$$Outlook, \quad Humidity, \quad Wind$$

Highest IG:

$$Outlook = 0.247$$

Therefore:

$$\boxed{Outlook = \text{Root}}$$

Step 2: Create the first branches

$$Outlook \rightarrow \begin{cases} Sunny \\ Overcast \\ Rain \end{cases}$$

Overcast is already pure:

$$S\{+4, -0\} = 0$$

Therefore:

$$Overcast \rightarrow Yes$$

Step 3: Continue with Sunny

Sunny is:

$$S\{+2, -3\}$$

It is not pure.

Calculate IG using the remaining features:

$$IG(Humidity \mid Sunny) = 0.971$$

$$IG(Wind \mid Sunny) = 0.020$$

Therefore:

$$Sunny \rightarrow Humidity$$

Then:

$$Humidity = High \rightarrow No$$

$$Humidity = Normal \rightarrow Yes$$

Step 4: Continue with Rain

Rain is:

$$S\{+3, -2\}$$

It is not pure.

Calculate IG using the remaining features:

$$IG(Humidity \mid Rain) = 0.020$$

$$IG(Wind \mid Rain) = 0.971$$

Therefore:

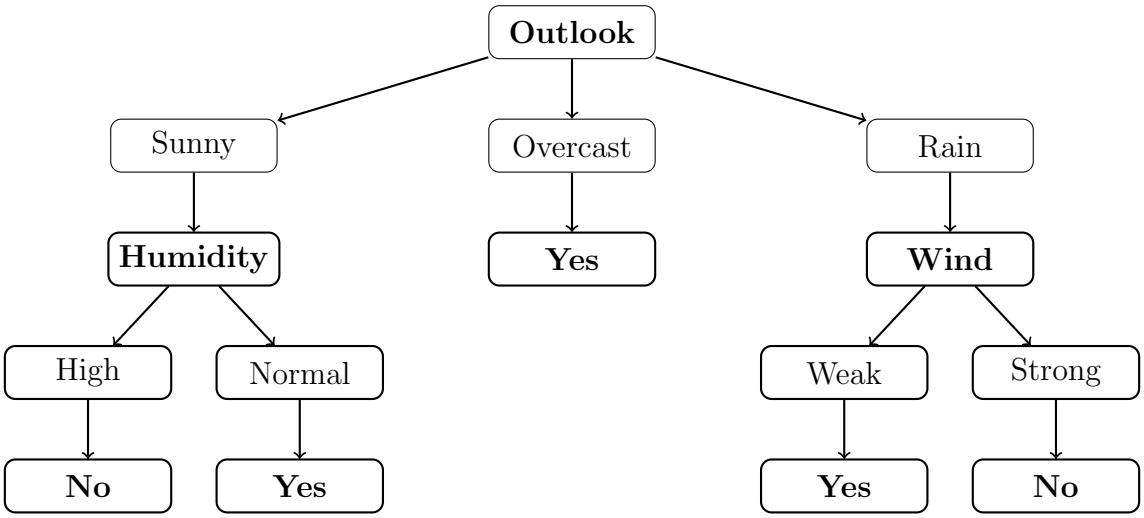
$$Rain \rightarrow Wind$$

Then:

$$Wind = Weak \rightarrow Yes$$

$$Wind = Strong \rightarrow No$$

17 17. Final Decision Tree



18 18. How to Read the Final Tree

The tree can be read as a sequence of decisions.

1. First check the **Outlook**.
2. If Outlook is **Overcast**:
$$\boxed{Play\ Football = Yes}$$
3. If Outlook is **Sunny**, check Humidity:
   - High → No
   - Normal → Yes
4. If Outlook is **Rain**, check Wind:
   - Weak → Yes
   - Strong → No

Thus, the Decision Tree converts the dataset into a set of simple decision rules.

19 19. Important Points to Remember

Key Concept

1. Start with the complete dataset.
2. Write the target count as:
$$S\{+x, -y\}$$
3. Calculate entropy of the current dataset.
4. Calculate Information Gain for every available feature.
5. The feature with the highest IG becomes the next decision node.
6. If a branch contains only one class, stop and create a leaf.
7. If a branch is still mixed, repeat the same process using the remaining features.
8. Continue until the tree is constructed.